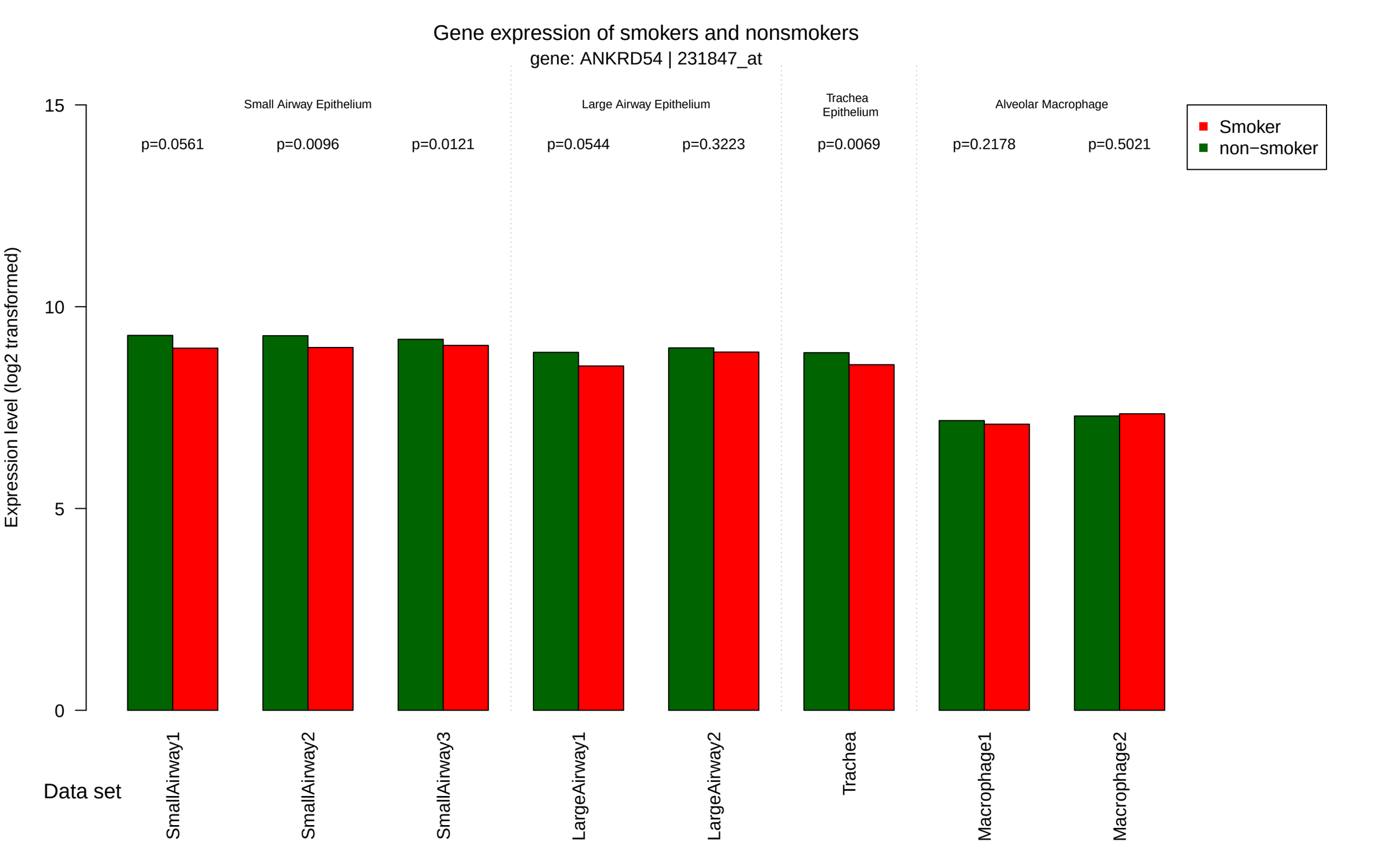
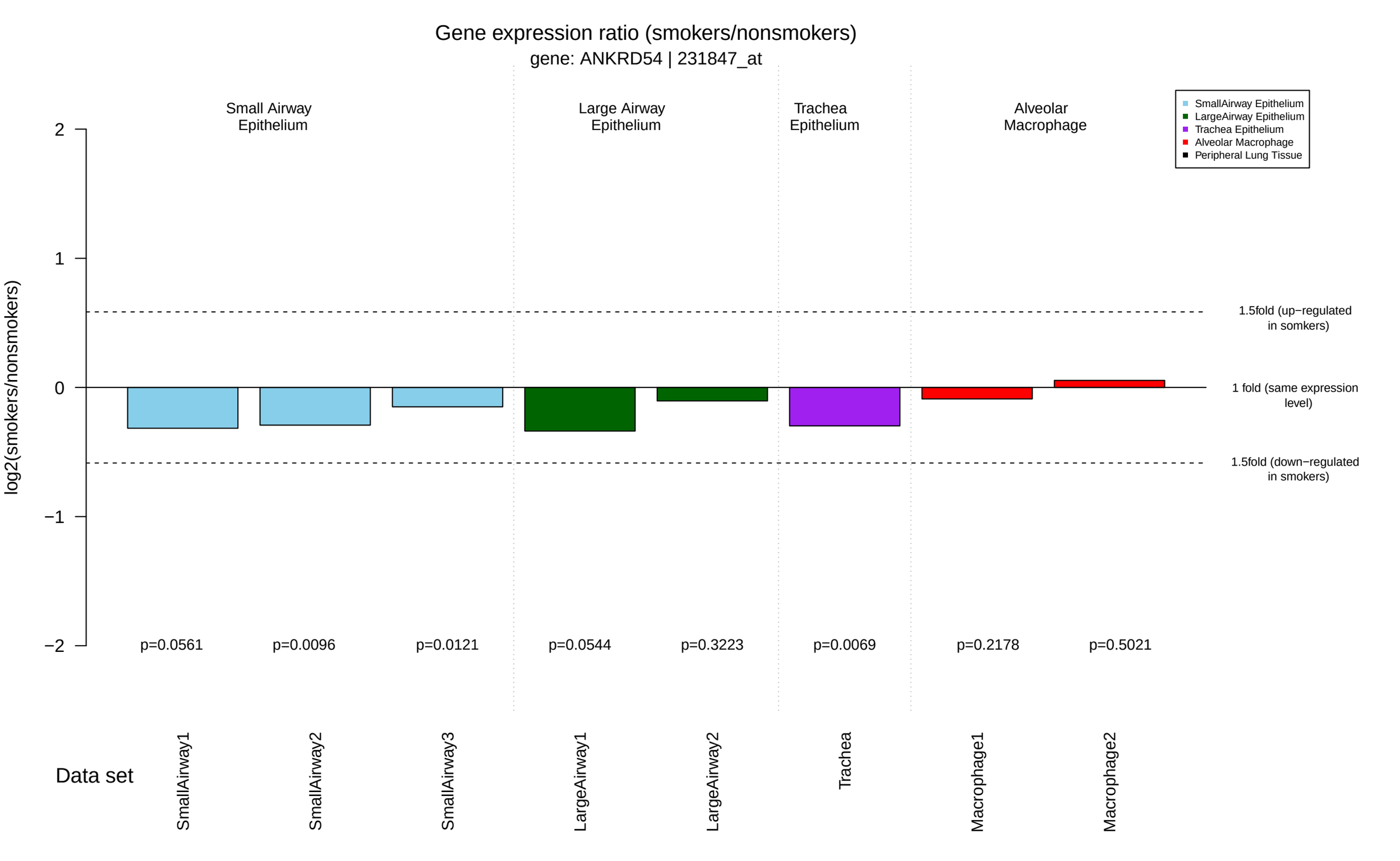
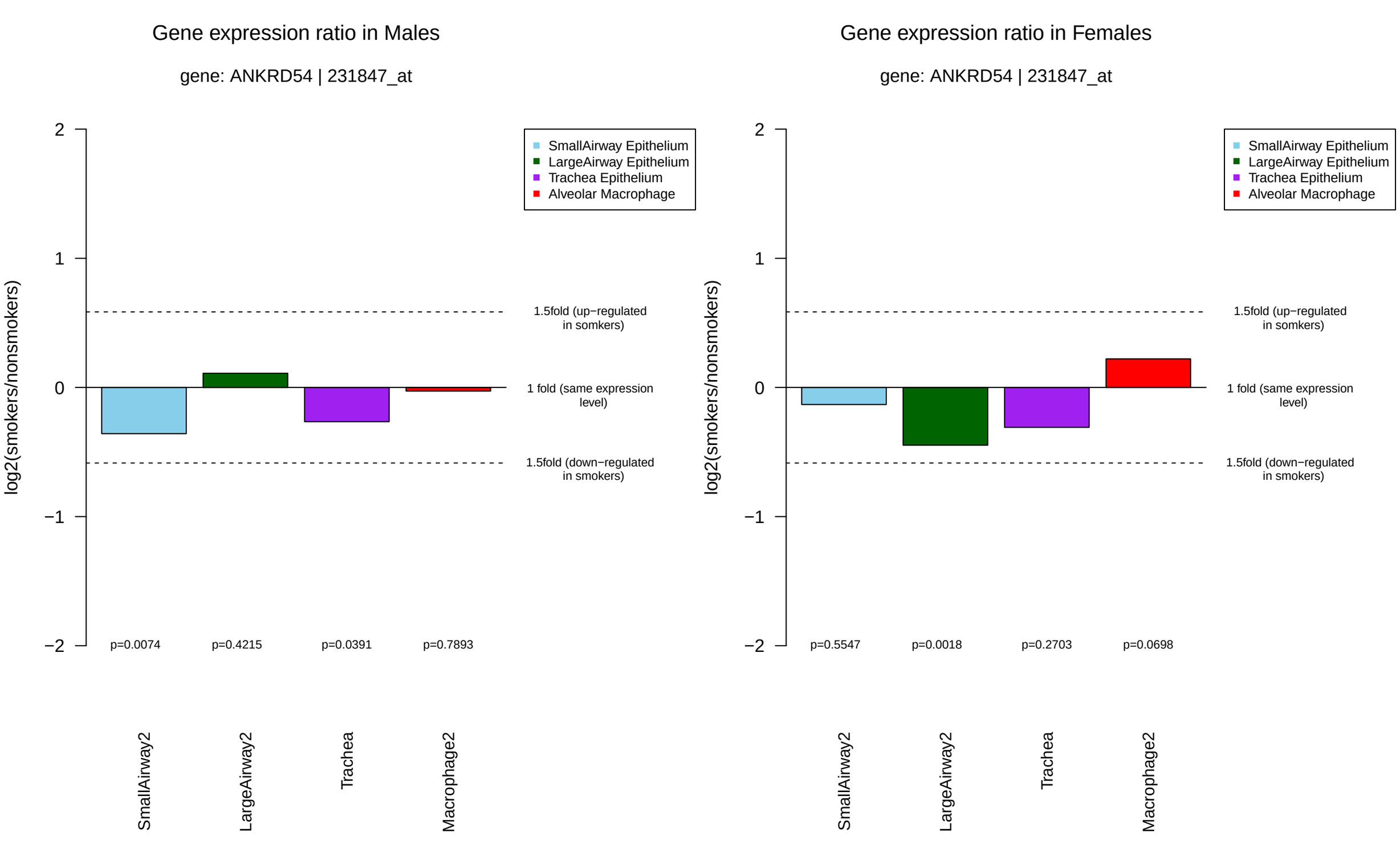
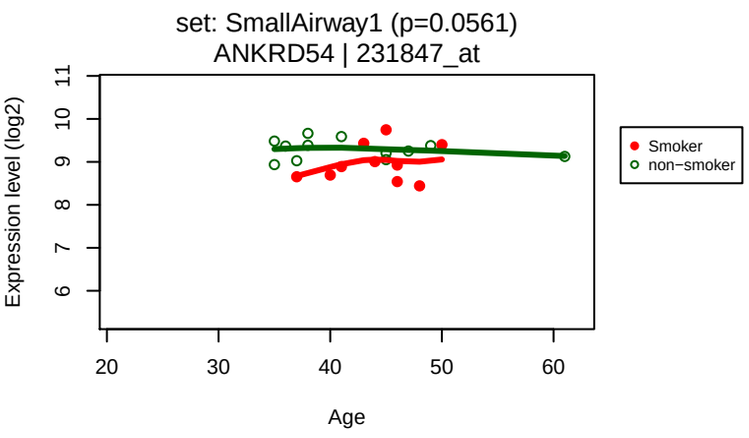
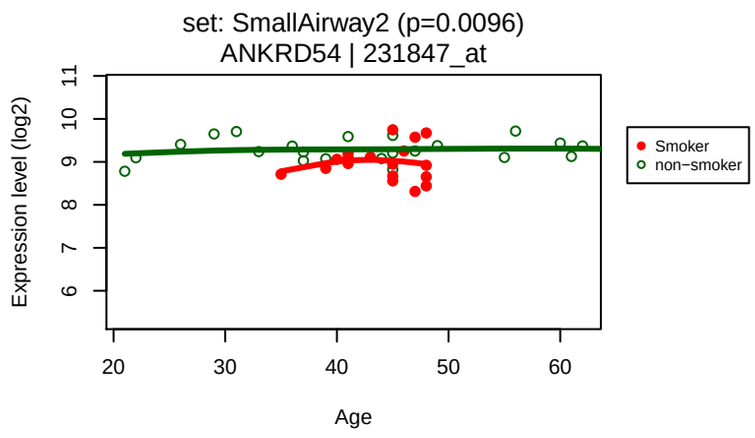
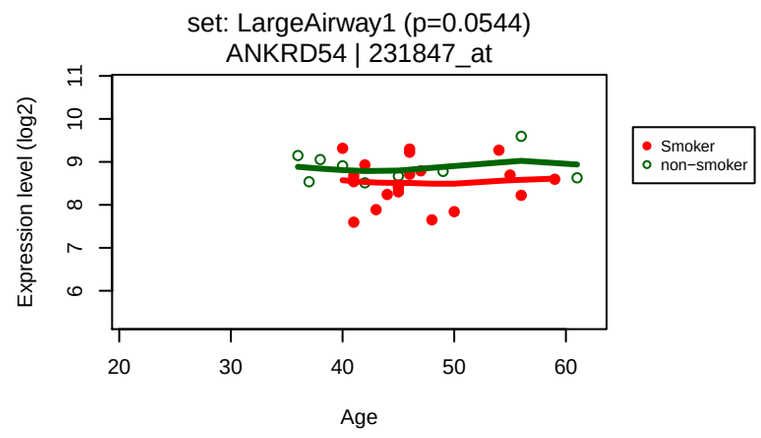

Gene expression of smokers and nonsmokers
gene: ANKRD54 | 231847_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0561
p=0.0096
p=0.0121
p=0.0544
p=0.3223
p=0.0069
p=0.2178
p=0.5021
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

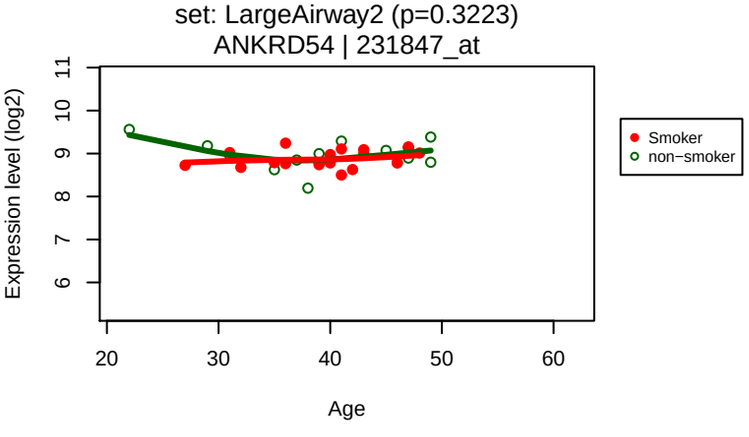

Gene expression ratio (smokers/nonsmokers)
gene: ANKRD54 | 231847_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0561
p=0.0096
p=0.0121
p=0.0544
p=0.3223
p=0.0069
p=0.2178
p=0.5021
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

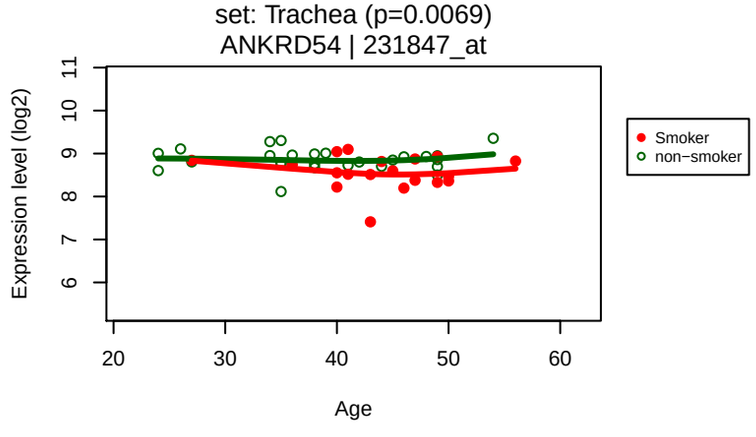

Gene expression ratio in Males
gene: ANKRD54 | 231847_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0074
p=0.4215
p=0.0391
p=0.7893
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ANKRD54 | 231847_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5547
p=0.0018
p=0.2703
p=0.0698
SmallAirway2
LargeAirway2
Trachea
Macrophage2

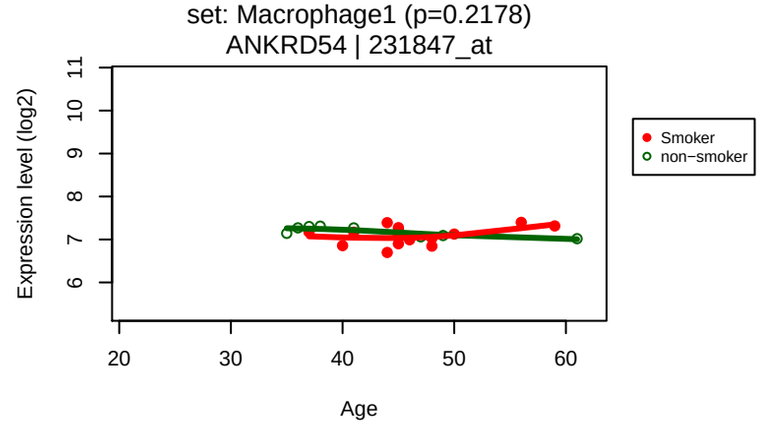

set: SmallAirway1 (p=0.0561)
ANKRD54 | 231847_at
Expression level (log2)
Age
Smoker
non−smoker

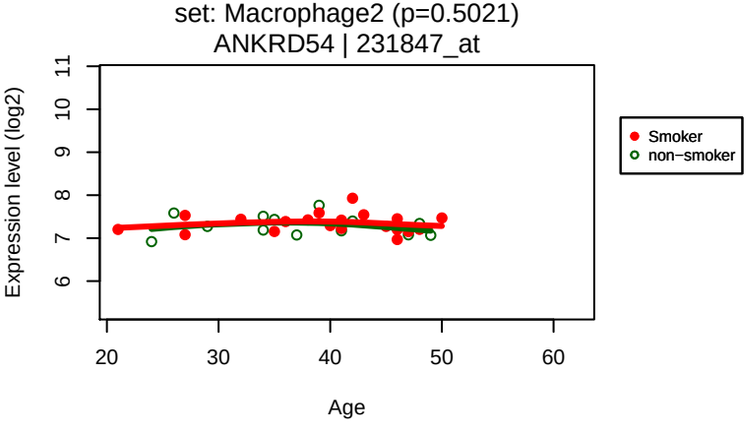

set: SmallAirway2 (p=0.0096)
ANKRD54 | 231847_at
Expression level (log2)
Age
Smoker
non−smoker

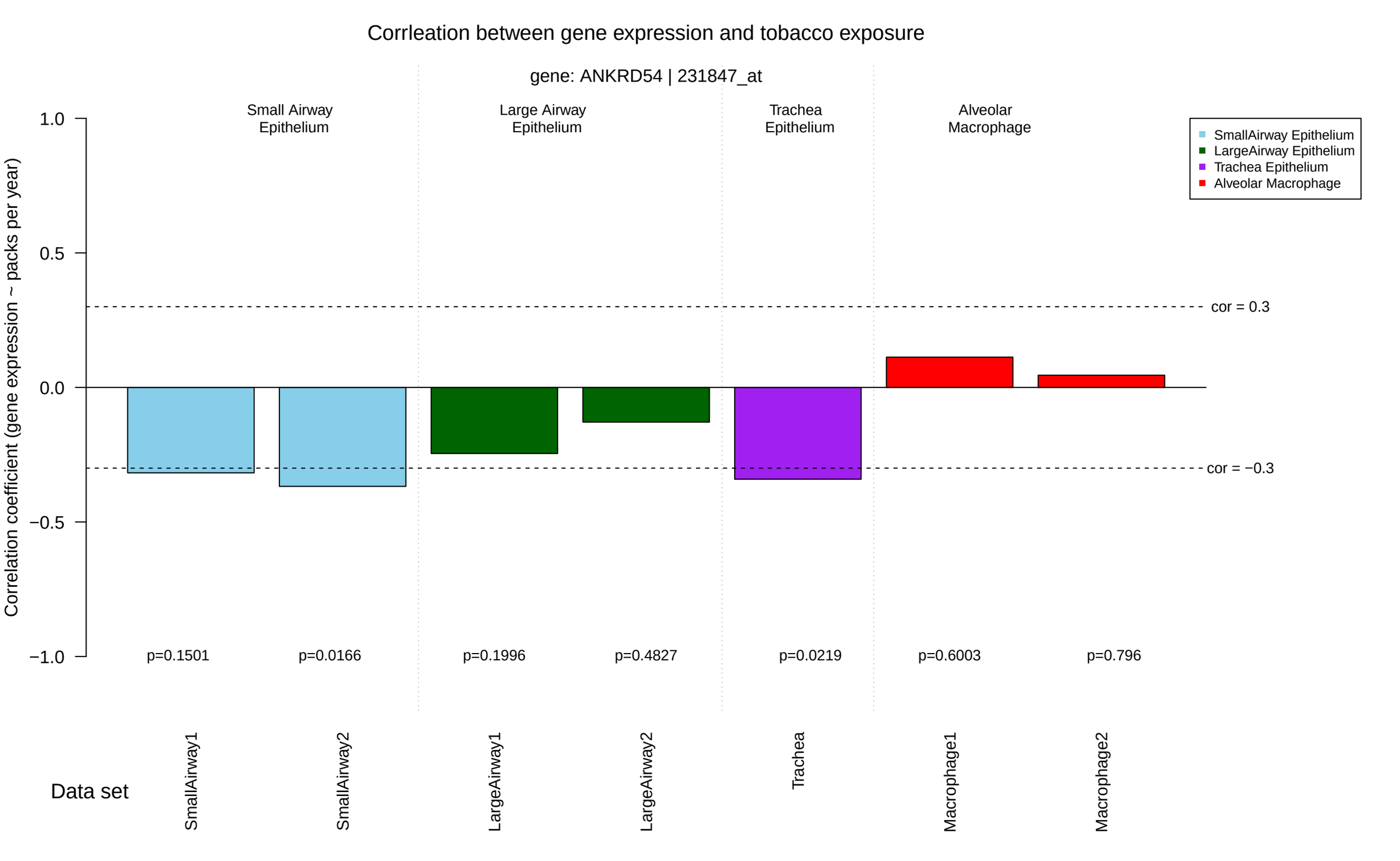

set: LargeAirway1 (p=0.0544)
ANKRD54 | 231847_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3223)
ANKRD54 | 231847_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0069)
ANKRD54 | 231847_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2178)
ANKRD54 | 231847_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5021)
ANKRD54 | 231847_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ANKRD54 | 231847_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1501
p=0.0166
p=0.1996
p=0.4827
p=0.0219
p=0.6003
p=0.796
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2